

Problem set 1

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1a $\sum_{k=1}^3 k^2 = 1^2 + 2^2 + 3^2 = 1 + 4 + 9 = \boxed{14}$

b $\sum_{k=1}^3 a_k^2 = \boxed{a_1^2 + a_2^2 + a_3^2}$

c $\sum_{j=1}^3 \sum_{k=1}^3 a_{jk} b_{kj} = \sum_{j=1}^3 \left[\sum_{k=1}^3 a_{jk} b_{kj} \right]$

$$= \sum_{j=1}^3 a_{j1} b_{1j} + a_{j2} b_{2j} + a_{j3} b_{3j}$$

$$= \boxed{a_{11} b_{11} + a_{21} b_{12} + a_{31} b_{13} + a_{12} b_{21} + a_{22} b_{22} + a_{32} b_{23} + a_{13} b_{31} + a_{23} b_{32} + a_{33} b_{33}}$$

d $\left(\sum_{j=1}^3 a_j \right)^2 = \sum_{j=1}^3 \sum_{k=1}^3 a_j a_k = \sum_{j=1}^3 \left[\sum_{k=1}^3 a_j a_k \right] =$

$$\sum_{j=1}^3 a_j a_1 + a_j a_2 + a_j a_3$$

$$= \boxed{a_1 a_1 + a_2 a_1 + a_3 a_1 + a_1 a_2 + a_2 a_2 + a_3 a_2 + a_1 a_3 + a_2 a_3 + a_3 a_3}$$

2. $v_x = 1, v_y = 2, v_z = -5$ $w_x = 3, w_y = -1, w_z = 1$

a) $(v \cdot w) = (1 \cdot 3) + (2 \cdot -1) + (-5 \cdot 1) = 3 - 2 - 5 = \boxed{-4}$

b) $[v \times w] = ((2 \cdot 1) - (-5 \cdot -1))i + ((-5 \cdot 3) - (1 \cdot 1))j + ((1 \cdot -1) - (2 \cdot 3))k$
 $\quad \quad \quad \boxed{-3i + -16j + -7k}$

c) The length of $v =$
 $\quad \quad \quad \sqrt{1^2 + 2^2 + (-5)^2} = \sqrt{1+4+25} = \boxed{\sqrt{30}}$

d) $(\delta_x \cdot v) = \boxed{1}$

e) $[\delta_x \times w] = \begin{vmatrix} \delta_x & \delta_y & \delta_z \\ 1 & 0 & 0 \\ 3 & -1 & 1 \end{vmatrix} = \delta_x((0 \cdot 1) - (0 \cdot -1)) + \delta_y((0 \cdot 3) - (1 \cdot 1)) + \delta_z((1 \cdot -1) - (0 \cdot 3)) = \boxed{\delta_y(-1) + \delta_z(-1)}$

f) $\phi_{vw} = \cos \theta = \frac{v \cdot w}{vw} = \frac{-4}{\sqrt{11} \cdot \sqrt{30}} = \frac{-4}{\sqrt{330}} = -0.220 = \boxed{77^\circ}$

$w = \sqrt{3^2 + (-1)^2 + 1^2} = \sqrt{11}$

g) $(r \times v)$, where r is the position vector

$= \begin{vmatrix} \delta_x & \delta_y & \delta_z \\ x & y & z \\ 1 & 2 & -5 \end{vmatrix} = \boxed{\delta_x(-5y - 2z) + \delta_y(z + 5x) + \delta_z(2x - y)}$

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$$1. \quad \begin{array}{lll} T_{xx} = 3 & T_{xy} = 2 & T_{xz} = -1 \\ T_{yx} = 2 & T_{yy} = 2 & T_{yz} = 1 \\ T_{zx} = -1 & T_{zy} = 1 & T_{zz} = 4 \end{array} \quad v_x = 5, v_y = 3, v_z = -2$$

a) $[\mathbf{T} \cdot \mathbf{v}] =$

$$\delta_x: (3 \cdot 5) + (2 \cdot 3) + (-1 \cdot -2) = 15 + 6 + 2 = 23$$

$$\delta_y: (2 \cdot 5) + (2 \cdot 3) + (1 \cdot -2) = 10 + 6 + -2 = 14$$

$$\delta_z: (-1 \cdot 5) + (1 \cdot 3) + (4 \cdot -2) = -5 + 3 + -8 = -10$$

$$\boxed{\delta_x(23) + \delta_y(14) + \delta_z(-10)}$$

b) $[\mathbf{v} \cdot \mathbf{T}] =$

$$\delta_x: (5 \cdot 3) + (3 \cdot 2) + (-2 \cdot -1) = 23$$

$$\delta_y: (5 \cdot 2) + (3 \cdot 2) + (-2 \cdot 1) = 14$$

$$\delta_z: (5 \cdot -1) + (3 \cdot 1) + (-2 \cdot 4) = -10$$

$$\boxed{\delta_x(23) + \delta_y(14) + \delta_z(-10)}$$

$$c) (\mathbf{T} : \mathbf{T}) = \sum_{i=1}^3 \sum_{j=1}^3 T_{ij} T_{ji} = \sum_i \sum_j T_{ij}^2 = 3^2 + 2^2 + (-1)^2 + 2^2 + 2^2 + 1^2 + (-1)^2 + 1^2 + 4^2$$

$$9 + 4 + 1 + 4 + 4 + 1 + 1 + 1 + 16$$

$$20 + 10 + 10 + 1 = \boxed{41}$$

d) $(\mathbf{v} \cdot [\mathbf{T} \cdot \mathbf{v}]) = (5 \cdot 23) + (3 \cdot 14) + (-2 \cdot -10) = \boxed{177}$

e) $\mathbf{v}\mathbf{v} =$

$$5(5, 3, -2) = 25, 15, -10$$

$$3(5, 3, -2) = 15, 9, -6$$

$$-2(5, 3, -2) = -10, -6, 4$$

$$\boxed{\begin{bmatrix} 25 & 15 & -10 \\ 15 & 9 & -6 \\ -10 & -6 & 4 \end{bmatrix}}$$

$$f) [\mathbf{T} \cdot \delta_1] =$$

$$\delta x: (3 \cdot 1) + 0 + 0 = 3$$

$$\delta y: (2 \cdot 1) + 0 + 0 = 2$$

$$\delta z: (-1 \cdot 1) + 0 + 0 = -1$$

$$\boxed{\delta x(3) + \delta y(2) + \delta z(-1)}$$

$$2a) [\delta_1 \delta_2 \cdot \delta_2] \times \delta_1$$

$$(\delta_2 \cdot \delta_2) (\delta_1 \times \delta_1) = \boxed{0}$$

$$b) (\delta : \delta_1 \delta_2) =$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} : \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \boxed{0}$$

$$c) (\delta : \delta)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} : \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 1+1+1 = \boxed{3}$$

$$d) \{\delta \cdot \delta\} =$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \boxed{8}$$

$$x: 1, 0, 0$$

$$y: 0, 1, 0$$

$$z: 0, 0, 1$$

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1. evaluate $(\nabla \cdot v)$, ∇v and $[\nabla \cdot vv]$

a) $v_x = by$ $v_y = 0$ $v_z = 0$

$$\nabla \cdot v = 0$$

$$\nabla v = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix} = \begin{bmatrix} 0 & b & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\nabla \cdot vv = (\nabla \cdot v)v + (v \cdot \nabla)v$$

$$(\nabla \cdot v)v = 0 \cdot v = 0$$

$$(v \cdot \nabla)v = \boxed{0, 0, 0}$$

b) $v_x = bx$ $v_y = 0$ $v_z = 0$

$$\nabla \cdot v = \frac{\partial(bx)}{\partial x} + \frac{\partial(0)}{\partial y} + \frac{\partial(0)}{\partial z} = b + 0 + 0 = \boxed{b}$$

$$\nabla v = \begin{bmatrix} b & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \frac{\partial v_x}{\partial x} = b$$

$$(\nabla \cdot v)v = (b)(bx, 0, 0) = (b^2x, 0, 0)$$

$$(v \cdot \nabla)v = (b)(bx, 0, 0) = (b^2x, 0, 0) = \boxed{(2b^2x, 0, 0)}$$

c) $v_x = by$ $v_y = bx$ $v_z = 0$

$$\nabla \cdot v = \frac{\partial(by)}{\partial x} + \frac{\partial(bx)}{\partial y} + \frac{\partial(0)}{\partial z} = \boxed{0}$$

$$\nabla v = \begin{bmatrix} 0 & b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \frac{\partial(v_x)}{\partial y} = b, \quad \frac{\partial(v_y)}{\partial x} = b$$

$$(\nabla \cdot v)v = (0)(v) = 0$$

$$(v \cdot \nabla)v = \boxed{(b^2x, b^2y, 0)}$$

d) $v_x = -by$ $v_y = bx$ $v_z = 0$

$$(\nabla \cdot \mathbf{v}) = 0$$

$$\nabla \mathbf{v} = \begin{bmatrix} 0 & -b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(\mathbf{v} \cdot \nabla) \mathbf{v} = \left(-by \frac{\partial}{\partial x} + bx \frac{\partial}{\partial y} \right) \mathbf{v}$$

$$x: bx(-b) = -b^2x$$

$$y: -by(b) = -b^2y$$

$$\boxed{(-b^2x, -b^2y, 0)}$$

9. If $\mathbf{r}$ is the position vector and r is its magnitude, verify that

a) $\nabla \frac{1}{r} = \frac{-\mathbf{r}}{r^3}$ $\mathbf{r} = (x, y, z)$
 $r = \sqrt{x^2 + y^2 + z^2}$

$$\frac{1}{r} = (x^2 + y^2 + z^2)^{-\frac{1}{2}}$$

$$\frac{\partial}{\partial x} \left(\frac{1}{r} \right) = -\frac{1}{2} (x^2 + y^2 + z^2)^{-\frac{3}{2}} \cdot (2x) = -\frac{x}{r^3}$$

$$\frac{\partial}{\partial y} \left(\frac{1}{r} \right) = -\frac{y}{r^3}$$

$$\frac{\partial}{\partial z} \left(\frac{1}{r} \right) = -\frac{z}{r^3}$$

$$\nabla \left(\frac{1}{r} \right) = \left(-\frac{x}{r^3}, -\frac{y}{r^3}, -\frac{z}{r^3} \right) = \left(-\frac{1}{r^3} \right) (x, y, z) = \frac{-\mathbf{r}}{r^3} \checkmark$$

c. $\nabla(\mathbf{a} \cdot \mathbf{r}) = \mathbf{a}$

$$\mathbf{a} \cdot \mathbf{r} = a_x x + a_y y + a_z z$$

$$\frac{\partial}{\partial x} (a_x x + a_y y + a_z z) = a_x$$

$$\frac{\partial}{\partial y} = a_y$$

$$\nabla(\mathbf{a} \cdot \mathbf{r}) = (a_x, a_y, a_z) = \mathbf{a} \checkmark$$

$$\frac{\partial}{\partial z} = a_z$$

$$b) \nabla f(r) = \frac{1}{r} \frac{df}{dr} r$$

$$\nabla f(r) = \frac{df}{dr} \cdot \left(\frac{\partial r}{\partial x}, \frac{\partial r}{\partial y}, \frac{\partial r}{\partial z} \right)$$

$$\frac{\partial r}{\partial x} = \frac{1}{2}(x^2 + y^2 + z^2)^{\frac{1}{2}} (2x) = \frac{x}{r}$$

$$\frac{\partial r}{\partial y} = \frac{y}{r}$$

$$\frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\nabla f(r) = \left(\frac{df}{dr} \right) \left(\frac{x}{r}, \frac{y}{r}, \frac{z}{r} \right) = \frac{1}{r} \frac{df}{dr} r \quad \checkmark$$

$$10a) \frac{\partial}{\partial t} p v = -[\nabla \cdot p v v] - \nabla p - [\nabla \cdot \tau] + \rho g$$

$$\frac{\partial}{\partial t} p v_x = - \left(\frac{\partial}{\partial x} p v_x v_x + \frac{\partial}{\partial y} p v_y v_x + \frac{\partial}{\partial z} p v_z v_x \right) - \frac{\partial p}{\partial x} - \left(\frac{\partial}{\partial x} \tau_{xx} + \frac{\partial}{\partial y} \tau_{yx} + \frac{\partial}{\partial z} \tau_{zx} \right) + \rho g_x$$

$$\frac{\partial}{\partial t} p v_y = - \left(\frac{\partial}{\partial x} p v_x v_y + \frac{\partial}{\partial y} p v_y v_y + \frac{\partial}{\partial z} p v_z v_y \right) - \frac{\partial p}{\partial y} - \left(\frac{\partial}{\partial x} \tau_{xy} + \frac{\partial}{\partial y} \tau_{yy} + \frac{\partial}{\partial z} \tau_{zy} \right) + \rho g_y$$

$$\frac{\partial}{\partial t} p v_z = - \left(\frac{\partial}{\partial x} p v_x v_z + \frac{\partial}{\partial y} p v_y v_z + \frac{\partial}{\partial z} p v_z v_z \right) - \frac{\partial p}{\partial z} - \left(\frac{\partial}{\partial x} \tau_{xz} + \frac{\partial}{\partial y} \tau_{yz} + \frac{\partial}{\partial z} \tau_{zz} \right) + \rho g_z$$

$\begin{matrix} x & y & z \\ yx & xy & yz \\ zx & zy & zz \end{matrix}$

b) $T = -\mu(\nabla v + (\nabla v)^T - \frac{2}{3}(\nabla \cdot v)\delta)$

$$T_{xx} = -\mu \left(2 \frac{\partial v_x}{\partial x} \right) - \frac{2}{3} \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right)$$

$$T_{yy} = -\mu \left(2 \frac{\partial v_y}{\partial y} \right) - \frac{2}{3} \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right)$$

$$T_{zz} = -\mu \left(2 \frac{\partial v_z}{\partial z} \right) - \frac{2}{3} \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right)$$

$$T_{xy} = T_{yx} = -\mu \left(\frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y} \right) - \frac{2}{3} \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right)$$

$$T_{xz} = T_{zx} = -\mu \left(\frac{\partial v_z}{\partial x} + \frac{\partial v_x}{\partial z} \right) - \frac{2}{3} \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right)$$

$$T_{yz} = T_{zy} = -\mu \left(\frac{\partial v_z}{\partial y} + \frac{\partial v_y}{\partial z} \right) - \frac{2}{3} \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right)$$

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1. $V = \delta_1 x_1 + \delta_2 x_3 + \delta_3 x_2$

$$x_1 = 0, x_1 = 1, x_2 = 0, x_2 = 2, x_3 = 0, x_3 = 4$$

$$\int_V (\nabla \cdot v) dV = \int_S (n \cdot v) dS$$

$$\nabla \cdot v = \frac{\partial x_1}{\partial x_1} + \frac{\partial x_3}{\partial x_2} + \frac{\partial x_2}{\partial x_3} = 1 + 0 + 0 = 1$$

$$\int_0^4 \int_0^2 \int_0^1 (1) dx dy dz = (1)(2)(4) = \boxed{8}$$

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1. $r = r\delta_r + z\delta_z$

$$v = \frac{d}{dt}(r\delta_r + z\delta_z)$$

$$v = \dot{r}\delta_r + r\dot{\delta}_r + \dot{z}\delta_z = \dot{r}\delta_r + \dot{\theta}\delta_\theta + \dot{z}\delta_z \checkmark$$

$$\dot{\delta}_r = \frac{d}{dt}(\delta_x \cos \theta + \delta_y \sin \theta)$$

$$= \delta_x \frac{d}{dt} \cos \theta \frac{d\theta}{dt} + \delta_y \frac{d}{dt} \sin \theta \frac{d\theta}{dt} = \dot{\theta} \delta_\theta$$

$$a = \frac{dv}{dt} = \frac{d}{dt} (\delta_r \dot{r} + \delta_\theta \dot{\theta} + \delta_z \dot{z})$$

$$\frac{d}{dt} S_\theta = \frac{d}{dt} (-\delta_x \sin \theta + \delta_y \cos \theta) = -\delta_x (\cos \theta) \dot{\theta} - \delta_y (\sin \theta) \dot{\theta} = -\delta_r \dot{\theta}$$

$$\begin{aligned} \frac{dv}{dt} &= \dot{S}_r \dot{r} + \delta_r \ddot{r} + \dot{S}_\theta \dot{\theta} + \delta_\theta \ddot{\theta} + \delta_\theta r \ddot{\theta} + \delta_z \ddot{z} \\ &= (\delta_\theta \dot{\theta}) \dot{r} + \delta_r \ddot{r} + (-\delta_r \dot{\theta}) r \dot{\theta} + \delta_\theta \ddot{\theta} + \delta_\theta r \ddot{\theta} + \delta_z \ddot{z} \\ &= \boxed{\delta_r (\ddot{r} - r \dot{\theta}^2) + \delta_\theta (r \ddot{\theta} + 2 \dot{r} \dot{\theta}) + \delta_z \ddot{z}} \end{aligned}$$

3a) $(\nabla \cdot p\mathbf{v})$, where p is a scalar

$$(\nabla \cdot p\mathbf{v}) = \frac{1}{r} \frac{\partial}{\partial r} (rp v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (p v_\theta) + \frac{\partial}{\partial z} (p v_z)$$

b) $[\nabla \cdot p\mathbf{v}]_r$, where p is a scalar

$$[\nabla \cdot p\mathbf{v}]_r = \frac{1}{r} \frac{\partial}{\partial r} (rp v_r^2) + \frac{1}{r} \frac{\partial}{\partial \theta} (p v_\theta v_r) + \frac{\partial}{\partial z} (p v_z v_r) - \frac{p v_\theta^2}{r}$$

$$c) [\nabla \cdot p\delta]_\theta = \frac{1}{r} \frac{\partial}{\partial r} (rp) - \frac{1}{r}$$

$$d) (\nabla \cdot [\boldsymbol{\tau} \cdot \mathbf{v}]) = \frac{1}{r} \frac{\partial}{\partial r} (r (\tau_{rr} v_r + \tau_{r\theta} v_\theta + \tau_{rz} v_z)) + \frac{1}{r} \frac{\partial}{\partial \theta} (\tau_{\theta r} v_r + \tau_{\theta\theta} v_\theta + \tau_{\theta z} v_z) + \frac{\partial}{\partial z} (\tau_{zr} v_r + \tau_{z\theta} v_\theta + \tau_{zz} v_z)$$

$$e) [\mathbf{v} \cdot \nabla \mathbf{v}]_\theta = v_r \left(\frac{\partial v_\theta}{\partial r} \right) + v_\theta \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) + v_z \left(\frac{\partial v_z}{\partial z} \right)$$

$$f) \nabla \mathbf{v} + (\nabla \mathbf{v})^+ =$$

$$[\nabla \mathbf{v} + (\nabla \mathbf{v})^+]_{rr} = 2 \left(\frac{\partial v_r}{\partial r} \right)$$

$$[\nabla \mathbf{v} + (\nabla \mathbf{v})^+]_{\theta\theta} = 2 \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right)$$

$$a) \operatorname{div}(p \langle v |) = \langle \nabla p | v \rangle + p \operatorname{div} \langle v |$$

$$\nabla \cdot (p v) = \frac{\partial (p v_i)}{\partial x_i}$$

$$\frac{\partial (p v_i)}{\partial x_i} = v_i \frac{\partial p}{\partial x_i} + p \frac{\partial v_i}{\partial x_i}$$

$$= \langle \nabla p | v \rangle + \nabla \cdot v = \langle \nabla p | v \rangle + p \operatorname{div} \langle v | \checkmark$$

$$b) \operatorname{div} \langle v | v \rangle \langle w | = \langle \nabla v | w \rangle + \langle \operatorname{div} w | v \rangle$$

$$(v w)_{ij} = v_i w_j$$

$$(\nabla \cdot (v w))_i = \frac{\partial (v_i w_j)}{\partial x_j}$$

$$\frac{\partial v_i w_j}{\partial x_j} = w_j \frac{\partial v_i}{\partial x_j} + v_i \frac{\partial w_j}{\partial x_j}$$

$$= \langle \nabla v | w \rangle + \langle \operatorname{div} w | v \rangle \checkmark$$

$$c) \operatorname{div} \langle v | \tau | = |\tau| : |\nabla v| + \langle v | \operatorname{div} \tau \rangle$$

$$\langle v | \tau |_{ij} = v_i \tau_{ij}$$

$$\operatorname{div} \langle v | \tau | = \frac{\partial}{\partial x_j} (v_i \tau_{ij})$$

$$\text{Product Rule: } \tau_{ij} \frac{\partial v_i}{\partial x_j} + v_i \frac{\partial \tau_{ij}}{\partial x_j}$$

$$|\tau| : |\nabla v| + \langle v | \operatorname{div} \tau \rangle \checkmark$$